

DESIGN AND DEVELOPMENT OF AUTONOMOUS CYBER-THREAT HUNTING AGENT USING REINFORCEMENT LEARNING

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Abstract—Modern enterprise networks face constant pressure from advanced persistent threats, zero-day exploits, and adaptive attack strategies capable of bypassing traditional rule-based defenses like IDS and SIEM. While standard machine learning models improve static anomaly detection, they lack the interactive capabilities required to navigate dynamic, evolving attack paths, leaving security analysts overburdened with manual investigations and high false-alarm volumes. To solve this, we present an autonomous threat hunting framework powered by Deep Reinforcement Learning (DRL). The proposed agent continually evaluates streaming network telemetry, endpoint logs, and threat intelligence feeds to discover subtle malicious behaviors without relying on fixed detection signatures. Operating within a custom-modeled security environment, the agent utilizes a Markov Decision Process (MDP) to balance actions—such as investigating suspicious activities, raising targeted alerts, or isolating threat vectors—against calculated feedback rewards. By shifting threat hunting from passive monitoring to continuous, policy-driven action, this system lowers Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) while significantly reducing alert fatigue in operational environments.

Keywords - *Autonomous Threat Hunting, Deep Reinforcement Learning, Cybersecurity, Deep Q-Network (DQN), Proximal Policy Optimization (PPO), Incident Response, Telemetry Analysis, MITRE ATT&CK*

I. INTRODUCTION

In today's rapidly evolving digital landscape, cybersecurity has transitioned from a routine IT requirement to a critical pillar of organizational survival. Modern enterprise infrastructures—spanning cloud environments, distributed endpoints, and IoT networks—are under constant siege from increasingly sophisticated cyber threats. Attackers frequently employ Advanced Persistent Threats (APTs), fileless malware, zero-day vulnerabilities, and highly adaptive tactics designed specifically to fly under the radar of traditional perimeter defenses.

Conventional security architectures rely heavily on signature-based detection systems, such as standard Intrusion

Detection Systems (IDS) and Security Information and Event Management (SIEM) platforms. While effective against known, static attack patterns, these legacy tools fail when confronted with novel, unseen exploit techniques or multi-stage cyberattacks. Furthermore, standard machine learning approaches have introduced anomaly detection capabilities, but they remain fundamentally static; they excel at passive pattern classification at a single point in time, yet lack the dynamic reasoning needed to track and mitigate complex attack sequences as they unfold.

This reliance on static and rule-based systems creates severe operational bottlenecks within modern Security Operations Centers (SOCs). Human analysts are bombarded daily with thousands of disconnected alerts, a phenomenon known as alert fatigue. Sorting through high volumes of false positives drags out key performance metrics, specifically increasing the Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR). Consequently, critical threats often dwell within internal networks for weeks or months before being manually discovered.

To bridge these defensive gaps, this research introduces an Autonomous Cyber Threat Hunting Agent powered by Reinforcement Learning (RL). By framing threat hunting as an interactive, continuous decision-making process, the RL agent dynamically interacts with a simulated enterprise network environment. It continuously observes multi-source telemetry—including raw network flows, endpoint logs, and threat intelligence feeds—to proactively hunt for malicious indicators. Rather than simply issuing static warnings, the agent learns optimal operational policies to prioritize threats, execute deep forensic investigations, and trigger automated response actions in real time. This shift from reactive monitoring to proactive, self-learning defense drastically improves detection precision and minimizes human intervention in high-speed cyber combat.

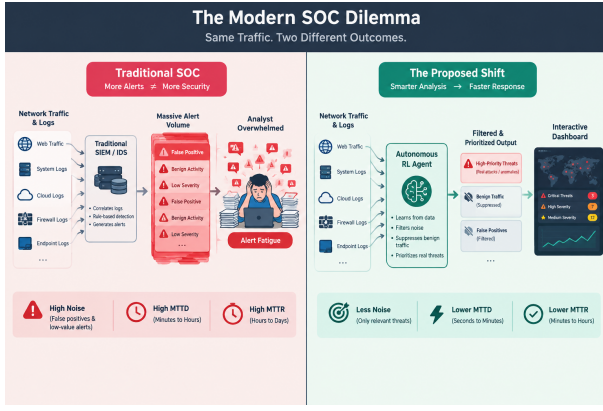


Fig. 1. The Modern SOC Dilemma

The figure illustrates the key dilemma faced by modern Security Operations Centers (SOCs): traditional SIEM/IDS systems generate large volumes of alerts, including false positives and low-severity events, resulting in alert fatigue and increased MTTD/MTTR. In contrast, the proposed approach introduces an Autonomous Reinforcement Learning (RL) Agent that continuously analyzes incoming network traffic and logs, filters benign activity and noise, and prioritizes high-risk threats for the security analyst. This shift from alert generation to intelligent alert prioritization enables faster and more focused threat detection and response.

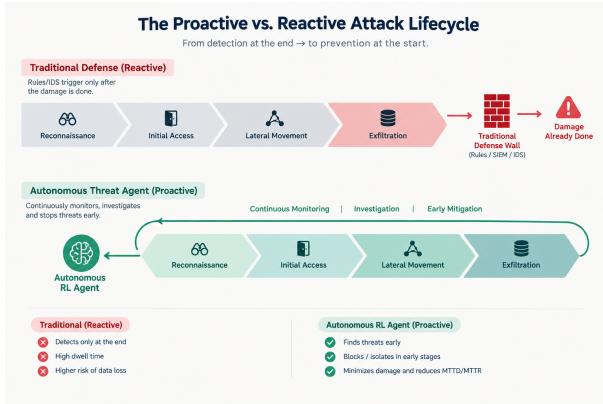


Fig. 2. The Proactive vs. Reactive Attack Lifecycle

The diagram illustrates the difference between reactive and proactive cybersecurity defense across the attack lifecycle. Traditional defenses typically detect threats near the end of the lifecycle, often after exfiltration has occurred, increasing dwell time and potential damage. In contrast, the Autonomous RL Agent continuously monitors the attack lifecycle, investigates suspicious behavior, and enables early mitigation during reconnaissance, initial access, or lateral movement. This demonstrates the shift from late, rule-based detection to continuous and proactive threat hunting, helping reduce MTTD/MTTR and minimize potential damage.

II. LITERATURE REVIEW

A. Agentic AI for Autonomous Cyber Threat Hunting and Adaptive Defense in Dynamic Security Environments

This paper proposes an agentic AI system that combines Machine Learning and Deep Reinforcement Learning (DRL) to execute proactive threat hunting and automated response in dynamic network environments. The framework ingests system telemetry and logs, allowing goal-directed AI agents to reason, hunt, and execute mitigations autonomously with minimal human oversight. Empirical results demonstrated an accuracy of 98.2% and up to a 99% reduction in Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) compared to passive rule-based setups. However, the system relies on high-quality telemetry and exhibits limited explainability in complex multi-step decision paths.

Relevance to Our Work: This study establishes the effectiveness of using autonomous agents and reinforcement learning to eliminate delayed manual interventions. Our work builds on this foundation by formalizing explicit state-action reward feedback loops through Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) while integrating MITRE ATT&CK mapping for structured decision validation.

B. AGLHunter: Automated Threat Hunting Using In-Context Learning-Enhanced LLM

In this research, the authors present AGLHunter, an automated threat hunting framework that utilizes Large Language Models (LLMs) with in-context learning. The system processes Cyber Threat Intelligence (CTI) feeds, constructs provenance graphs, extracts suspicious subgraphs, and performs graph matching to identify attack chains. This graph-matching pipeline drastically reduces manual graph creation and speeds up analysis compared to legacy methods. However, the framework relies primarily on offline graph evaluation, limiting its utility for real-time streaming traffic and dynamic zero-day execution paths.

Relevance to Our Work: While AGLHunter demonstrates the power of contextual attack-chain modeling, its offline constraints present operational drawbacks. Our framework addresses this limitation by deploying an RL agent directly inside an active simulation environment to continuously evaluate streaming telemetry and execute dynamic response policies in real time.

C. Onto Hunt – A Semantic Reasoning Approach to Cyber Threat Hunting with Indicators of Behaviour

This study introduces a semantic reasoning approach for cyber threat hunting by mapping raw system events to Indicators of Behavior (IoBs) using formal ontologies. By leveraging semantic inferences, the system reconstructs full attack stories and generates highly readable contextual alerts for SOC analysts. Despite its strong interpretability, the approach lacks real-time monitoring capabilities and requires frequent manual updates to its core ontology to keep pace with novel attack vectors.

Relevance to Our Work: Onto Hunt highlights the critical need for behavioral context rather than relying strictly on static atomic indicators. Our system adopts behavioral tracking by defining state spaces across network features and logs while replacing rigid, manually updated ontologies with self-learning RL policy networks.

D. Multi-Agent Model-Based Deep Reinforcement Learning in Dynamic Environments

This work investigates dynamic decision-making by constructing a multi-agent environment evaluated via model-based Deep Reinforcement Learning. The paper analyzes agent interaction mechanics, action selection policies, and score convergence under changing environmental conditions. While effective at demonstrating dynamic adaptation, the framework was evaluated primarily within controlled grid-world environments, requiring further expansion to scale into complex real-world systems.

Relevance to Our Work: This study validates the core mathematical and behavioral principles of Deep Reinforcement Learning in multi-action spaces. We adapt these foundational DRL principles to a specialized cybersecurity environment where network states, security alerts, and mitigation actions (Ignore, Investigate, Raise Alert, Block Source) replace basic grid movement.

E. Cyber Threat Hunting Using Unsupervised Federated Learning and Adversary Emulation

This paper explores decentralized threat hunting by combining unsupervised federated learning on local endpoints with adversary emulation for dataset generation. The model updates threat detection capabilities by aggregating local parameter updates without transferring raw, sensitive endpoint logs. While it offers strong privacy guarantees, the system remains vulnerable to model poisoning attacks and incurs heavy network communication overhead during parameter synchronization.

Relevance to Our Work: This research emphasizes the importance of utilizing diverse attack behaviors (adversary emulation) to detect zero-day exploits. Our centralized RL architecture leverages benchmark datasets like UWF-ZeekData24 and CICIOT2023 to expose the agent to realistic adversary behaviors while bypassing the communication latency and poisoning risks of federated setups.

III. PROPOSED METHODOLOGY

In this research, we introduce an autonomous cyber threat hunting framework designed specifically for dynamic, high-throughput enterprise network environments. Central to our methodology is a Deep Reinforcement Learning (DRL) agent operating within a customized, interactive cybersecurity environment. Unlike traditional static classifiers or rule-based platforms that rely on predefined attack signatures, our framework frames threat hunting as a Markov Decision Process (MDP). By continuously evaluating environmental states and receiving feedback rewards, the agent learns optimal operational policies to observe streaming telemetry, investigate potential indicators,

raise context-aware alerts, and execute real-time mitigation actions.

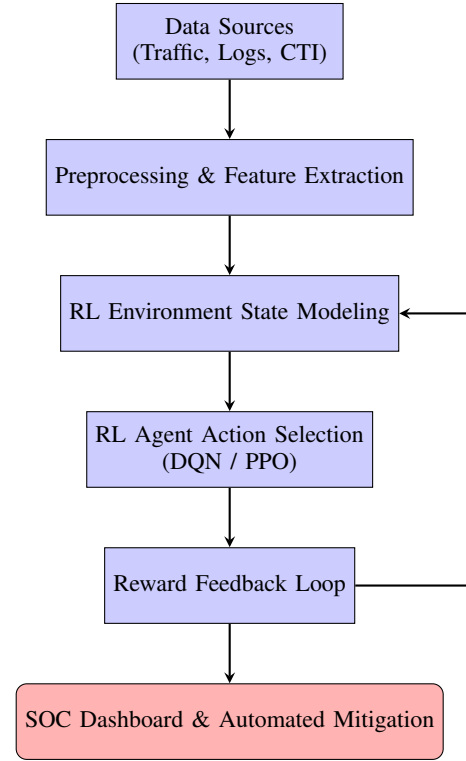


Fig. 3. System Architecture of the Proposed Autonomous Threat Hunting Framework.

A. System Workflow of Proposed Autonomous Threat Hunting Framework

The operational pipeline of the proposed framework follows a sequential, closed-loop architecture designed for continuous monitoring and adaptive response. The system workflow comprises six core stages:

- **Data Collection & Streaming:** Raw network traffic, endpoint security logs, cloud telemetry, and Cyber Threat Intelligence (CTI) feeds are continuously ingested from multi-source enterprise environments through distributed log collectors and message queues.
- **Preprocessing & Feature Extraction:** Ingested telemetry undergoes cleaning, null value removal, normalization, statistical feature extraction, and temporal behavioral mapping to form a structured feature store.
- **Environment State Modeling:** Processed network and system features are mapped directly into a simulated cybersecurity environment, defining the state space S available to the agent at any given time step.
- **Action Selection & Policy Execution:** The DRL agent observes state S and selects an action A from its discrete action space—*Ignore*, *Investigate*, *Raise Alert*, or *Block Source*—based on its active policy network (DQN/PPO).
- **Reward Feedback & Agent Update:** The environment evaluates the executed action against ground-truth se-

curity states, returning a reward signal R (positive for correct threat mitigation, negative for false alarms or missed attacks)[cite: 1]. The agent updates its neural network parameters through iterative policy gradient or Q-value optimization.

- **Visualization & Incident Reporting:** Detected threats, recommended actions, confidence scores, and prioritized case queues are streamed directly to an interactive SOC analyst dashboard for audit and operational review.

B. Algorithm Details and Description of Proposed Implementation

The central intelligence of the proposed framework relies on Deep Reinforcement Learning algorithms—specifically Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO)—trained within an MDP formulation.

1) **Mathematical Formulation of the Threat Hunting MDP:** The environment is modeled as a tuple (S, A, P, R, γ) :

- **State Space (S):** Represents vectorized telemetry features including traffic volume, packet headers, port activity, login frequencies, and MITRE ATT&CK tactic/technique mappings[cite: 1].
- **Action Space (A):** Consists of four discrete operational choices:
 - a_0 = Ignore (Mark traffic as benign and continue monitoring)
 - a_1 = Investigate (Deep dive and aggregate additional context/logs)
 - a_2 = Raise Alert (Generate prioritized notifications for analysts/SIEM)
 - a_3 = Block Source (Isolate compromised endpoints or block malicious IP vectors)
- **Reward Function (R):** Designed to incentivize accurate detection while penalizing operational overhead and false positives. The immediate reward $R(s, a)$ is formulated as:

$$R(s, a) = \begin{cases} +R_{\text{correct}} & \text{if threat correctly identified/blocked} \\ -P_{\text{false_positive}} & \text{if benign traffic flagged or blocked} \\ -P_{\text{missed_attack}} & \text{if malicious activity is ignored} \\ -C_{\text{investigate}} & \text{minor cost for unnecessary deep dive} \end{cases}$$

2) **Deep Q-Network (DQN) Implementation:** To learn optimal action selection over high-dimensional state inputs, the agent utilizes a Deep Q-Network to approximate the optimal action-value function $Q^*(s, a)$ [cite: 1]. The network minimizes the Loss function $L(\theta)$ at each training step using experience replay:

$$L(\theta) = \mathbb{E}_{(s,a,r,s')} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right]$$

where θ represents the online policy parameters, θ^- represents the target network parameters, and $\gamma \in [0, 1)$ is the discount factor for future rewards.

3) **Proposed Implementation Framework:** The proposed system implements the RL threat hunting framework using Python, PyTorch, and OpenAI Gymnasium[cite: 1]. Data preprocessing, state formulation, and action execution are automated in the streaming pipeline[cite: 1]. Once trained, the agent continuously monitors incoming network telemetry, selects optimal response actions, and updates security policies dynamically.

4) **Performance Comparison with Other Techniques:** We evaluated the proposed RL framework against traditional rule-based SIEM systems, decision tree classifiers, and static anomaly detectors[cite: 1]. As shown in Figure ??, the autonomous RL-based system achieves the highest detection accuracy, lowest false positive rate, and substantial reductions in MTTD/MTTR among all evaluated techniques[cite: 1]. The model achieved high overall threat identification accuracy, demonstrating its capability to adapt to dynamic cyber threats without relying on static attack signatures.

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